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Tensors and Differential Forms

1.1 Multilinear Algebra

Definition 1.1.1

Write $L(V_1, \dots, V_k; W)$ for the set of all multilinear maps from $V_1 \times \dots \times V_k$ to W . It is a vector space under the usual operations of pointwise addition and scalar multiplication:

$$(F + F')(v_1, \dots, v_k) = F(v_1, \dots, v_k) + F'(v_1, \dots, v_k),$$

$$(aF)(v_1, \dots, v_k) = a(F(v_1, \dots, v_k)).$$

Definition 1.1.2

Let $V_1, \dots, V_k, W_1, \dots, W_l$ be real vector spaces, and suppose $F \in L(V_1, \dots, V_k; \mathbb{R})$ and $G \in L(W_1, \dots, W_l; \mathbb{R})$. Define a function

$$F \otimes G : V_1 \times \dots \times V_k \times W_1 \times \dots \times W_l \rightarrow \mathbb{R}$$

by

$$F \otimes G(v_1, \dots, v_k, w_1, \dots, w_l) = F(v_1, \dots, v_k)G(w_1, \dots, w_l). \quad (1.1)$$

$F \otimes G$ is an element of $L(V_1, \dots, V_k, W_1, \dots, W_l; \mathbb{R})$, called the **tensor product of F and G** .

Proposition 1.1.3 ▶ A Basis for the Space of Multilinear Functions

Let $V_1, \dots, V_k$ be real vector spaces of dimensions $n_1, \dots, n_k$, respectively. For each $j \in \{1, \dots, k\}$, let $(E_1^{(j)}, \dots, E_{n_j}^{(j)})$ be a basis for V_j , and let $(\varepsilon_{(j)}^1, \dots, \varepsilon_{(j)}^{n_j})$ be the corresponding dual basis for V_j^* . Then the set

$$\mathcal{B} = \left\{ \varepsilon_{(1)}^{i_1} \otimes \dots \otimes \varepsilon_{(k)}^{i_k} : 1 \leq i_1 \leq n_1, \dots, 1 \leq i_k \leq n_k \right\}$$

is a basis for $L(V_1, \dots, V_k; \mathbb{R})$, which therefore has dimension equal to $n_1 \cdots n_k$.

Proof. • **$\mathcal{B}$ spans $L(V_1, \dots, V_k; \mathbb{R})$:** Suppose $F \in L(V_1, \dots, V_k; \mathbb{R})$ is arbitrary. Define $F_{i_1 \dots i_k} = F(E_{i_1}^{(1)}, \dots, E_{i_k}^{(k)})$. Then for any k -tuples of vectors $(v_1, \dots, v_k) \in V_1 \times \cdots \times V_k$, we write $v_1 = v_1^{i_1} E_{i_1}^{(1)}, \dots, v_k = v_k^{i_k} E_{i_k}^{(k)}$, then

$$\begin{aligned} F(v_1, \dots, v_k) &= v_1^{i_1} \cdots v_k^{i_k} F(E_{i_1}^{(1)}, \dots, E_{i_k}^{(k)}) \\ &= v_1^{i_1} \cdots v_k^{i_k} F_{i_1 \dots i_k} \\ &= F_{i_1 \dots i_k} \varepsilon_{(1)}^{i_1} \otimes \cdots \otimes \varepsilon_{(k)}^{i_k} (v_1, \dots, v_k) \end{aligned}$$

Thus

$$F = F_{i_1 \dots i_k} \varepsilon_{(1)}^{i_1} \otimes \cdots \otimes \varepsilon_{(k)}^{i_k}$$

• **$\mathcal{B}$ are linearly independent:** If the linear combination of $\mathcal{B}$, say $F_{i_1 \dots i_k} \varepsilon_{(1)}^{i_1} \otimes \cdots \otimes \varepsilon_{(k)}^{i_k}$ is equal to zero, by applying it on $(E_{j_1}^{(1)}, \dots, E_{j_k}^{(k)})$, we get $F_{j_1} \cdots F_{j_k} = 0$.

□

Covariant and Contravariant Tensors on a Vector Space

Definition 1.1.4

Let V be a finite-dimensional real vector space. If k is a positive integer, a **covariant k -tensor on V** is an element of the k -fold tensor product $V^* \otimes \cdots \otimes V^*$, which we typically think of as a real-valued multilinear function of k elements of V :

$$\alpha : \underbrace{V \times \cdots \times V}_{k \text{ copies}} \rightarrow \mathbb{R}.$$

1.2 Symmetric and Alternating Tensors

Symmetric Tensors

Definition 1.2.1 ▶ Symmetric Tensors

Let V be a finite-dimensional vector space. A covariant k -tensor α on V is said to be **symmetric** if its value is unchanged by interchanging any pair of arguments:

$$\alpha(v_1, \dots, v_i, \dots, v_j, \dots, v_k) = \alpha(v_1, \dots, v_j, \dots, v_i, \dots, v_k)$$

whenever $1 \leq i < j \leq k$.

Definition 1.2.2 ▶ $\Sigma^k(V^*)$

The set of symmetric covariant k -tensors is a linear subspace of the space $T^k(V^*)$ of all covariant k -tensors on V ; we denote this subspace by $\Sigma^k(V^*)$.

Definition 1.2.3 ▶ ${}^\sigma\alpha$

Given a k -tensor α and a permutation $\sigma \in S_k$, we define a new k -tensor ${}^\sigma\alpha$ by

$${}^\sigma\alpha(v_1, \dots, v_k) = \alpha(v_{\sigma(1)}, \dots, v_{\sigma(k)}).$$

Definition 1.2.4 ▶ Symmetrization

We define a projection

$$\text{Sym} : T^k(V^*) \rightarrow \Sigma^k(V^*)$$

called **symmetrization** by

$$\text{Sym } \alpha = \frac{1}{k!} \sum_{\sigma \in S_k} {}^\sigma\alpha.$$

More explicitly, this means that

$$(\text{Sym } \alpha)(v_1, \dots, v_k) = \frac{1}{k!} \sum_{\sigma \in S_k} \alpha(v_{\sigma(1)}, \dots, v_{\sigma(k)}).$$

Definition 1.2.5 ▶ symmetric product

If $\alpha \in \Sigma^k(V^*)$ and $\beta \in \Sigma^l(V^*)$, we define their **symmetric product** to be the $(k + l)$ -tensor $\alpha\beta$ (denoted by juxtaposition with no intervening product symbol) given by

$$\alpha\beta = \text{Sym}(\alpha \otimes \beta).$$

More explicitly, the action of $\alpha\beta$ on vectors $v_1, \dots, v_{k+l}$ is given by

$$\alpha\beta(v_1, \dots, v_{k+l}) = \frac{1}{(k+l)!} \sum_{\sigma \in S_{k+l}} \alpha(v_{\sigma(1)}, \dots, v_{\sigma(k)}) \beta(v_{\sigma(k+1)}, \dots, v_{\sigma(k+l)}).$$

Alternating Tensors**Definition 1.2.6 ▶ k -covectors**

We assume that V is a finite-dimensional real vector space. A covariant k -tensor α on V is said to be **alternating** (or **antisymmetric** or **skew-symmetric**) if it changes sign whenever two of its arguments are interchanged. This means that for all vectors $v_1, \dots, v_k \in V$ and every pair of distinct indices i, j it satisfies

$$\alpha(v_1, \dots, v_i, \dots, v_j, \dots, v_k) = -\alpha(v_1, \dots, v_j, \dots, v_i, \dots, v_k).$$

Alternating covariant k -tensors are also variously called **exterior forms**, **multi-vectors**, or **k -covectors**. The subspace of all alternating covariant k -tensors on V is denoted by $\Lambda^k(V^*) \subseteq T^k(V^*)$.

Definition 1.2.7

We define a projection $\text{Alt} : T^k(V^*) \rightarrow \Lambda^k(V^*)$, called **alternation**, as follows:

$$\text{Alt } \alpha = \frac{1}{k!} \sum_{\sigma \in S_k} (\text{sgn } \sigma)(\sigma\alpha),$$

where S_k is the symmetric group on k elements. More explicitly, this means

$$(\text{Alt } \alpha)(v_1, \dots, v_k) = \frac{1}{k!} \sum_{\sigma \in S_k} (\text{sgn } \sigma) \alpha(v_{\sigma(1)}, \dots, v_{\sigma(k)}).$$

1.3 The Algebra of Alternating Tensors

Definition 1.3.1

We assume on that V is a finite-dimensional real vector space. Given $\omega \in \Lambda^k(V^*)$ and $\eta \in \Lambda^l(V^*)$, we define their **wedge product** or **exterior product** to be the following $(k + l)$ -covector:

$$\omega \wedge \eta = \frac{(k + l)!}{k!l!} \text{Alt}(\omega \otimes \eta). \quad (1.2)$$

Remark.

系数的产生, 可以通过对 Alt 的另一种看法来理解:

- 先对所有 $(k + \ell)!$ 个排列求和.
- 但因为 α, β 已经交错, 同一个 (k, ℓ) -shuffle 等价类里的 $k! \ell!$ 个排列产生的值完全一样.
- 因此, Alt 本质上是

$$\frac{k! \ell!}{(k + \ell)!} \times (\text{对每个等价类选一个代表, 再对所有等价类求和}).$$

- 于是 wedge product 就等于

$$\alpha \wedge \beta = \frac{(k + \ell)!}{k! \ell!} \text{Alt}(\alpha \otimes \beta) = \sum_{\text{等价类代表}} \text{sgn}(\text{代表}) \cdot (\text{那一项}).$$

1.4 Tensors and Tensor Fields on Manifolds

1.5 Differential Forms on Manifolds

1.5.1 differential k-form

Definition 1.5.1 ► differential k-form

Let M be a n -dimensional smooth manifold (with or without boundary).

- The subset of $T^k T^* M$ consisting of alternant tensors is denoted by $\Lambda^k T^* M$:

$$\Lambda^k T^* M = \coprod_{p \in M} \Lambda^k (T_p^* M)$$

- A section of $\Lambda^k T^* M$ is called a **differential k-form**, or just a **k-form**. The integer k is called the **degree** of the form; this is a (continuous) tensor field whose value at each point is an alternating tensor.
- We denote the vector space of smooth k-forms by

$$\Omega^k(M) = \Gamma(\Lambda^k T^* M)$$

- If we define

$$\Omega^*(M) = \bigoplus_{k=0}^n \Omega^k(M)$$

then the wedge product turns $\Omega^*(M)$ into an associative, and it commutative graded algebra.

- In any smooth chart, a k-form ω can be written locally as

$$\omega = \sum_I \omega_I dx^{i_1} \wedge \cdots \wedge dx^{i_k} = \sum_I \omega_I dX^I$$

where the coefficients ω_I are continuous functions defined on the coordinate domain, and we use dx^I as an abbreviation for $dx^{i_1} \wedge \cdots \wedge dx^{i_k}$

Definition 1.5.2 ► pullback of differential forms

If $F : M \rightarrow N$ is a smooth map and ω is a differential form on N , then pullback $F^*\omega$ is a differential form on M , defined as for any covariant tensor field:

$$(F^*\omega)_p(v_1, \dots, v_k) = \omega_{F(p)}(dF_p(v_1), \dots, dF_p(v_k))$$

1.5.2 Exterior Derivatives**Definition 1.5.3**

Let $U \subseteq \mathbb{R}^n$ or $\mathbb{H}^n$, f is a 0-form on U , ω is a k -form on U . We define

•

$$df = \sum_{i=1}^n \frac{\partial f}{\partial x^i} dx^i$$

•

$$d\left(\sum_J \omega_J dx^J\right) = \sum_J d\omega_J \wedge dx^J$$

In smoeewhat more detail, this is

$$d\left(\sum_J \omega_J dx^{j_1} \wedge \dots \wedge dx^{j_k}\right) = \sum_J \sum_i \frac{\partial \omega_J}{\partial x^i} dx^i \wedge dx^{j_1} \wedge \dots \wedge dx^{j_k}$$

Proposition 1.5.4 ► Properties of the Exterior Derivative on $\mathbb{R}^n$

1. d is linear over $\mathbb{R}$.
2. If ω is a smooth k -form and η is a smooth l -form on an open subset $U \subseteq \mathbb{R}^n$ or $\mathbb{H}^n$, then

$$d(\omega \wedge \eta) = d\omega \wedge \eta + (-1)^k \omega \wedge d\eta.$$

3. $d \circ d \equiv 0$.
4. d commutes with pullbacks: if U is an open subset of $\mathbb{R}^n$ or $\mathbb{H}^n$, V is an open subset of $\mathbb{R}^m$ or $\mathbb{H}^m$, $F : U \rightarrow V$ is a smooth map, and $\omega \in \Omega^k(V)$, then

$$F^*(d\omega) = d(F^*\omega). \quad (1.3)$$

Theorem 1.5.5 ► Existence and Uniqueness of Exterior Differentiation

Suppose M is a smooth manifold with or without boundary. There are unique operators $d : \Omega^k(M) \rightarrow \Omega^{k+1}(M)$ for all k , called **exterior differentiation**, satisfying the following four properties:

- (i) d is linear over $\mathbb{R}$.
- (ii) If $\omega \in \Omega^k(M)$ and $\eta \in \Omega^l(M)$, then

$$d(\omega \wedge \eta) = d\omega \wedge \eta + (-1)^k \omega \wedge d\eta.$$

- (iii) $d \circ d \equiv 0$.
- (iv) For $f \in \Omega^0(M) = C^\infty(M)$, df is the differential of f , given by $df(X) = Xf$.

In any smooth coordinate chart, d is given by Theorem 1.5.3.

Definition 1.5.6

- If $A = \bigoplus_k A^k$ is a graded algebra, a linear map $T : A \rightarrow A$ is said to be a **map of degree m** if $T(A^k) \subseteq A^{k+m}$ for each k .
- It is said to be an **antiderivation** if it satisfies $T(xy) = (Tx)y + (-1)^k x(Ty)$ whenever $x \in A^k$ and $y \in A^l$.

Remark.

Differential on functions extends uniquely to an antiderivation of $\Omega^*(M)$ of degree +1 whose square is zero.

1.5.3 An Invariant Formula for the Exterior Derivative

Proposition 1.5.7 ► Exterior Derivative of a 1-Form

For any smooth 1-form ω and smooth vector fields X and Y ,

$$d\omega(X, Y) = X(\omega(Y)) - Y(\omega(X)) - \omega([X, Y]). \quad (1.4)$$

Proposition 1.5.8 ► Invariant Formula for the Exterior Derivative

Let M be a smooth manifold with or without boundary, and $\omega \in \Omega^k(M)$.

For any smooth vector fields $X_1, \dots, X_{k+1}$ on M ,

$$\begin{aligned} & d\omega(X_1, \dots, X_{k+1}) \\ &= \sum_{1 \leq i \leq k+1} (-1)^{i-1} X_i (\omega(X_1, \dots, \hat{X}_i, \dots, X_{k+1})) \\ &+ \sum_{1 \leq i < j \leq k+1} (-1)^{i+j} \omega([X_i, X_j], X_1, \dots, \hat{X}_i, \dots, \hat{X}_j, \dots, X_{k+1}), \end{aligned} \quad (1.5)$$

where the hats indicate omitted arguments.

1.5.4 Lie Derivatives of Differential Forms

Proposition 1.5.9

Suppose M is a smooth manifold, $V \in \mathfrak{X}(M)$, and $\omega, \eta \in \Omega^*(M)$. Then

$$\mathcal{L}_V(\omega \wedge \eta) = (\mathcal{L}_V \omega) \wedge \eta + \omega \wedge (\mathcal{L}_V \eta).$$

Theorem 1.5.10 ► Cartan's Magic Formula

On a smooth manifold M , for any smooth vector field V and any smooth differential form ω ,

$$\mathcal{L}_V \omega = V \lrcorner (d\omega) + d(V \lrcorner \omega). \quad (1.6)$$

Corollary 1.5.11 ► The Lie Derivative Commutes with d

If V is a smooth vector field and ω is a smooth differential form, then

$$\mathcal{L}_V(d\omega) = d(\mathcal{L}_V \omega).$$